De La Salle University - Dasmariñas

COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITCS318
COURSE TITLE	**:**	INFORMATION ASSURANCE AND SECURITY 1
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	2 Units
MODALITY	**:**	**Traditional Blended Learning**
PRE-REQUISITES	**:**	S-ITCS123, S-ITCS123LA
CO-REQUISITES	**:**	S-ITCS318LA
PROFESSOR	**:**	MARIVIC R. MITSCHEK
CONSULTATION HOURS	**:**	(M 800AM-1100AM-ONSITE)
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VISION

A leading Catholic university that inspires excellence and drives innovation towards a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension program.

COURSE DESCRIPTION:

This course examines fundamentals of network security involved in creating and managing secure computer network environments. Both hardware and software topics are considered including authentication methods, remote access, network security architectures and devices, cryptography, forensics, and disaster recovery plans. This course adopts the three courses included in the online curriculum of Junior Cybersecurity Analyst Path of the Cisco

Networking Academy including Introduction to Cybersecurity, Cybersecurity Threat Management and Endpoint Security.

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:

CLO1. Articulate the players in the cybersecurity world and the motivation of cyber criminals and cybersecurity specialists.
CLO2. Analyze security attacks, symptoms, processes, and countermeasures.
CLO3. Apply skills in security management, controls, protection, and mitigation technologies.
CLO4. Illustrate how cybersecurity professionals use technologies, processes and procedures to defend all components of the network.
CLO5. Create a security measure plan for the different Cybersecurity domains.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand the players in the cybersecurity world and the motivation of cyber criminals and cybersecurity specialists.	TLO1. Describe the common characteristics comprising the cybersecurity world TLO2. Understand the three principles of information assurance and security.
CLO2. Analyze security attacks, symptoms, processes, and countermeasures.	TLO3. Differentiate the types of malwares and malicious code. TLO4. Compare the different methods used in social engineering.
CLO3. Apply skills in security management, controls, protection, and mitigation technologies.	TLO5. Explain how encryption techniques protect confidentiality. TLO6. Explain processes used to ensure integrity. TLO7. Explain the concept of high availability.
CLO6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	Week	Teaching Learning Activities/Assessment Strategy	Technology Enabler	nsite/face-to-face	Asyncrhonous	Allotted Hours
Introduction Module	1 Aug 19-24	• Syllabus Uploaded • Gospel reading, prayer and reflection • Presentation of course introduction • Course requirement details	Schoolbook	✓		2
Module 1 TLO 1 CLO 1 CLO 6	2 Aug 26-31	• Introduction to Cybersecurity • Gospel reading, prayer and reflection • Enabling Assessment: o What was Taken? o Cybersecurity Job Hunt • Attacks, Concepts and Techniques	Schoolbook, Packet Tracer	✓		2
Module 2 TLO 2 CLO 1 CLO 6	3 Sept 2-7	• Protecting your Data and Privacy • Gospel reading, prayer and reflection • Enabling Assessment: o Who owns your data? • Protecting the Organization • Will Your Future be in Cybersecurity?	Schoolbook, Packet Tracer		✓	2
Module 1, 2, 3 TLO 2 CLO 1 CLO 6	4 Sept 9-14	**Summative Exam on Introduction to Cybersecurity Course** • Gospel reading, prayer and reflection • Enabling Assessment o Packet Tracer 4.3.3.3– Configuring VPN Transport o Packet Tracer 4.3.3.4– Configuring VPN Tunnel Mode	Schoolbook, Packet Tracer	✓		2
Module 4 TLO 3,4 CLO 1	5 Sept 16-21	• Governance and Compliance • Gospel reading, prayer and reflection • Enabling Assessment	Schoolbook, Packet Tracer		✓	2

CLO 6		• Developing Cybersecurity Policies and Procedures • Network Security Testing				
Module 5 TLO 5 CLO 2 CLO3 CLO 6	6 Sept 23-28	• Threat Intelligence • Gospel reading, prayer and reflection • Enabling Assessment o Lab – Risk Management o Lab – Risk Analysis o Lab – Security Controls Implementation	Schoolbook, Packet Tracer		✓	2
Module 6 TLO 6,7 CLO 2 CLO3 CLO 6	7 Sept 30 - Oct 5	• Digital Forensics & Incident Analysis & Response • Gospel reading, prayer and reflection • Lab – Attack Analysis • Lab – Incident Handling • Packet Tracer – Investigate Disaster Recovery	Schoolbook, Packet Tracer	✓		2
Module 4, 5, 6	8 Oct 7-12	**Cyber Threat Management Exam**	Schoolbook, Packet Tracer		✓	***2***
	9 Oct 14-19	**Midterm Exam**		✓		***2***
TOTAL				***10***	***8***	***18***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Apply skills in security management, controls, protection, and mitigation technologies.	TLO6. Explain processes used to ensure integrity. TLO7. Explain the concept of high availability.
CLO4. Explain how cybersecurity professionals use technologies, processes, and procedures to defend all components of the network.	TLO8. Explain how processes and procedures protect systems. TLO9. Explain how to implement security measures to protect network devices.

CLO5. Create a security measure plan for the different Cybersecurity domains.	TLO10. Document the security measure plan for the different Cybersecurity domains.
CLO6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching Learning Activities/Assessment Strategy**	**Technology Enabler**	**nsiteFace-to-face**	**Asynxhronous**	**Allotted Hours**
Module 7TLO 6 TLO 7 CLO 3 CLO 6	10 Oct 21-26	• Cybersecurity Threats, Vulnerabilities and Attacks • Gospel reading, prayer and reflection	Schoolbook, Packet Tracer	✓		2
	11 Oct 28 -Nov 2	**Independent Learning Days**				2
Module 7 TLO 6 TLO 7 CLO 3 CLO 6	12 Nov 4-9	• Cybersecurity Threats, Vulnerabilities and Attacks Gospel reading, prayer and reflection	Schoolbook, Packet Tracer	✓		2
Module 8 TLO 10 CLO 5 CLO 6	13 Nov 11-16	• Attacking the Foundation Gospel reading, prayer and reflection • Lab – Reading Server Logs Lab – Recommend Threat Mitigation Measures	Schoolbook, Packet Tracer		✓	2

Module 9 TLO 6 TLO 7 CLO 3 CLO 6	14 Nov18-23	• Wireless Network Communication Gospel reading, prayer and reflection Enabling Assessment o Configure Basic Wireless Security o Troubleshoot a Wireless Connection o Network Security Infrastructure o Packet Tracer – ACL Demonstration	Schoolbook, Packet Tracer		✓	2
Module 10 TLO 6 TLO 7 CLO 3 CLO 6	15 Nov 25-30	Summative Exam 1 **Checkpoint Exam**		✓		2
Module 11 TLO 8 TLO 9 CLO 4 CLO 6	16 Dec 2-7	• The Windows Operating System • Gospel reading, prayer and reflection • Enabling Assessment o Lab – Monitor and Manage System Resources in Windows Linux Overview	Schoolbook, Packet Tracer	✓		
Module 12 TLO 8 TLO 9 CLO 4 CLO 6	17 Dec 9-14	• System and Endpoint Protection • Gospel reading, prayer and reflection • Enabling Assessment o Lab – Harden a Linux System o Lab – Recover Passwords o Lab – Online Malware Investigation Tools o Cybersecurity Principles,	Schoolbook, Packet Tracer		✓	2

		Practices and Processes				
	18 Dec 16-18	**Summative assessment**		✓		2
			TOTAL	***10***	***8***	***18***

GRADING SYSTEM

Midterm

Enabling Assessments	- 50%
Class Participation	-20%
Summative Assessments	- 30%
	100%

Finals

Enabling Assessments	- 50%
Class Participation	-20%
Summative Assessments	- 30%
	100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.

b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:

i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Online References

Call number or eprovider	Reference Material
www.netacad.com	Cisco Networking Academy. [accessed : 2024 August 16].Retrieved from https://www.netacad.com/portal/resources/course-resources/cybersecurity-essentials
9781683924982	C. P. Gupta, & K. K. Goyal. (2020). *Cybersecurity : A Self-Teaching Introduction.* Mercury Learning & Information.
9781682824290	Hulick, K. (2019). *Cybersecurity Careers*. Reference Point Press.
10.1145/3527202	Wroclawski, J., & Benzel, T. (2022). *Cybersecurity as Illuminator for the Future of Computing Research: Considering the shifting fundamentals of cybersecurity research.* Communications of the ACM, 65(5), 39–41. https://doi.org/10.1145/3527202

Onsite References

Call number or eprovider	Reference Material
KF1263.C65 .K847 2019	Kosseff J. 2019. *Cybersecurity law*. Hoboken: Wiley.

Prepared: Information Technology Department
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